LESSON 6.5

Write and Evaluate Algebraic Expressions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an algebraic expression for a real situation and evaluate it for a given value of the variable.

WHY THIS MATTERS Evelyn is helping to plan the sixth-grade field day. The committee will order the items shown for each student participating in field day. They do not yet know how many students will be participating. The shipping cost for all items is a one-time cost of $23.50. What expression can Evelyn use to determine how much the committee will need to purchase the items?

REMEMBER

Write and Evaluate Algebraic Expressions

Writing & Evaluating Algebraic Expressions An algebraic expression uses numbers, a variable (a letter for an unknown number), and operations. We read a word phrase and write it with math symbols.

1. Define the variable to represent the unknown quantity.
2. Translate mathematical phrases into operations, coefficients, and constants.
3. Evaluate by substituting given numerical values for the variable

Words: **Variable *(Variable)*** **Algebraic Expression *(Expresión algebraica)*** **Coefficient *(Coeficiente)*** **Constant *(Constante)*** **Like terms *(Términos semejantes)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A square room has sides of 7 feet. Which expression gives its area, and what is the value?

(A) $2^7 = 128$
(B) $7^2 = 49$
(C) $7 \times 2 = 14$
(D) $7 + 7 = 14$

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A music studio charges a $25 setup fee plus $8 per hour of recording time. Let h represent the number of hours. Which expression models the total cost for h hours?

(A) 8 + 25h
(B) 25 + 8h
(C) 25h + 8h
(D) 33h

SHOW YOUR WORK

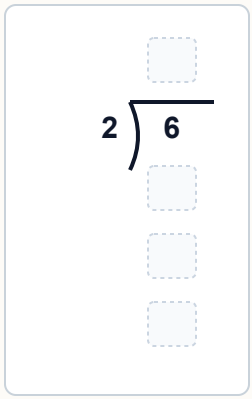
PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

In the expression 4 + 6 ÷ 2, which operation do you do first?

(A) Add 4 and 6
(B) Divide 6 by 2
(C) Either order gives the same answer
(D) Subtract



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

4 **Circle the letter of the best answer. Show how you know.**

Evaluate: $5 + 3 \times 2^2$

(A) 17
(B) 26
(C) 32
(D) 64

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

5 **Circle the letter of the best answer. Show how you know.**

Evaluate: $(8 - 3)^2 \div 5$

(A) 1
(B) 2.6
(C) 5
(D) 45

Rewrite the whole expression each line	What I simplified

1. **Grouping** *Inside ( ) first.*
2. **Exponents** *Then powers.*
3. **× ÷** *Left to right.*
4. **+ −** *Left to right.*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Phrase** the product of a number n and 4
2. **Student's expression** n + 4
3. **Reasoning** I saw two numbers and a variable, so I added them together.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Which operation does the key word 'product' name?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

SENTENCE STARTER The phrase '___ ' tells me to use ___ because the key word ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Line 1: A lyric says '7 more than a number of beats n.' Which expression matches?

(A) n + 7
(B) 7n
(C) n − 7
(D) 7 − n

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Line 2: The chorus says '5 less than a number of tracks x.' Which expression matches?

(A) x − 5
(B) 5 − x
(C) 5x
(D) x + 5

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

Line 3: A verse says 'twice a number of speakers' (use s). Which expression matches?

(A) 2s
(B) s + 2
(C) s − 2
(D) s ÷ 2

SHOW YOUR WORK

4 Write the name of the correct group on each line.

Sort each studio lyric by the operation its key word names.

Groups: Addition | Subtraction | Multiplication | Division

_____ the sum of a number of mics m and 9

_____ a number of riffs r increased by 6

_____ 4 less than a number of solos j

_____ a number of takes k decreased by 8

_____ the product of 5 and a number of loops g

_____ a number of verses v divided by 3

PART 2 Apply it

real-world problems

5 Circle the letter of the best answer. Show how you know.

A music studio charges a $25 setup fee plus $8 per hour of recording time. Let h represent the number of hours. What is the total cost for 6 hours?

(A) $33
(B) $48
(C) $73
(D) $198

PER 1

6 **Circle the letter of the best answer. Show how you know.**

A music studio charges a $25 setup fee plus $8 per hour of recording time. Let h represent the number of hours. In 25 + 8h, what is the coefficient?

(A) 25

(B) 8

(C) h

(D) 25h

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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